Calculus I — Entry Diagnostic Test

Course: Calculus of One Variable
Purpose: Assess prerequisite knowledge (algebra, precalculus, trigonometry)
Time allowed: 40 minutes
Instructions: Show all your work. No calculator allowed.

Problem 1 — Algebra and Functions

Tests factoring, rational expressions, and function composition — the core algebraic machinery used throughout calculus.

Let $f(x) = x^2 - 4$ and $g(x) = \dfrac{x+2}{x^2+x-6}$.

(a) Fully **factor** both $f(x)$ and the numerator and denominator of $g(x)$.

(b) Simplify $g(x)$ completely. State any values of x for which it is undefined.

(c) Compute and simplify $(f \circ g)(x)$, the composition of f with g, at $x = 4$.

Problem 2 — Graphs and Behavior of Functions

Tests whether students can read, interpret, and transform graphs — essential for understanding derivatives and integrals geometrically.

Consider the function $f(x) = -2(x-1)^2 + 8$.

(a) Identify the **vertex**, **axis of symmetry**, and direction of opening. Is this function even, odd, or neither?

(b) Find the **x-intercepts** and **y-intercept** algebraically.

(c) Sketch a **qualitative graph** of f, labeling the vertex and intercepts. Then describe what happens to the graph of f if it is transformed to $h(x) = -2(x-1)^2 + 8 + 3$ and to $k(x) = -2(2x-1)^2 + 8$.

Problem 3 — Trigonometry and Exponentials

Calculus of one variable relies heavily on trig identities and exponential/logarithm rules. This problem surfaces gaps in those areas.

(a) Given that $\sin(\theta) = \frac{3}{5}$ and θ is in the first quadrant, find $\cos(\theta)$ and $\tan(\theta)$ without a calculator.

(b) Simplify the following expression using trigonometric identities:

$$\frac{\sin^2(x) + \cos^2(x)}{\cos(x)} - \cos(x)$$

(c) Solve for x:

$$3^{2x-1} = 27$$

Then solve: $\ln(x+2) + \ln(3) = \ln(15)$.

Answer Key *(for instructor use)*

Problem 1

- **(a)** $f(x) = (x-2)(x+2)$; $g(x) = \dfrac{(x+2)}{(x+3)(x-2)}$.
- **(b)** $g(x) = \dfrac{1}{x+3}$, undefined at $x = 2$ and $x = -3$.
- **(c)** $g(4) = \dfrac{1}{7}$, then $f\left(\dfrac{1}{7}\right) = \dfrac{1}{49} - 4 = -\dfrac{195}{49}$.

Problem 2

- **(a)** Vertex $(1, 8)$; axis $x = 1$; opens downward. **Neither** (axis of symmetry not at $x = 0$).
- **(b)** x-intercepts: $-2(x-1)^2 = -8 \Rightarrow (x-1)^2 = 4 \Rightarrow x = 3$ or $x = -1$. y-intercept: $f(0) = -2 + 8 = 6$.
- **(c)** $h(x)$: vertical shift up 3 units. $k(x)$: horizontal compression by factor $\frac{1}{2}$.

Problem 3

- **(a)** By Pythagorean theorem: $\cos\theta = \dfrac{4}{5}$, $\tan\theta = \dfrac{3}{4}$.
- **(b)** $\dfrac{1}{\cos x} - \cos x = \dfrac{1 - \cos^2 x}{\cos x} = \dfrac{\sin^2 x}{\cos x} = \sin(x)\tan(x)$.
- **(c)** $3^{2x-1} = 3^3 \Rightarrow 2x - 1 = 3 \Rightarrow x = 2$. Logarithm: $\ln(3(x+2)) = \ln(15) \Rightarrow x + 2 = 5 \Rightarrow x = 3$.
